

ASSIGNMENT -03

$\mu = 55$, $\sigma_a = 4$, $\sigma_\beta = 10$, $\sigma_c = 15$ which is better?

Normal Distribution Comparison

Given: $\mu = 55$, $\sigma_a = 4$, $\sigma_\beta = 10$, $\sigma_c = 15$

Domain Chosen: Light Bulb Lifespan in Manufacturing Industry

Introduction

Statistics plays a crucial role in quality control and industrial manufacturing. One of the most important probability distributions used in industry is the **Normal Distribution (Gaussian Distribution)**. It is widely applied in:

- Quality control
- Product reliability testing
- Industrial production
- Risk management
- Engineering analysis

A normal distribution is defined by two parameters:

- **Mean (μ)** – average value
- **Standard Deviation (σ)** – variability or consistency

In this report, we analyze three normal distributions representing the lifespan of light bulbs manufactured in a factory.

Mean lifespan:

$\mu = 55$ days

Standard deviations:

$\sigma_a = 4$

$\sigma_\beta = 10$

$\sigma_c = 15$

All three cases have the same average lifespan but different variability. The objective is to determine which manufacturing process is better.

Mathematical Background

The probability density function (PDF) of a normal distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = mean lifespan
- σ = standard deviation
- e = Euler's number
- $\pi = 3.1416$

Important properties:

- Symmetric bell-shaped curve
- Mean = Median = Mode
- Total area under curve = 1
- Spread depends only on σ

Since $\mu = 55$ days is constant, only variability differs.

Understanding Mean ($\mu = 55$ days)

The mean represents the **average lifespan of bulbs**.

In all three cases:

Average lifespan = 55 days

This means:

- On average, bulbs last 55 days.
- The center of all distributions is the same.
- Customers expect roughly 55 days of usage.

Thus, the decision depends entirely on standard deviation.

Understanding Standard Deviation in Manufacturing

Standard deviation measures **consistency in production quality**.

Smaller σ :

- Bulbs last almost the same number of days
- High manufacturing precision
- High reliability
- Fewer customer complaints

Larger σ :

- Bulbs last very different lifespans
- Inconsistent production
- Some bulbs fail too early
- Customer dissatisfaction increases

Thus:

$\sigma = 4 \rightarrow$ Highly consistent production

$\sigma = 10 \rightarrow$ Moderate consistency

$\sigma = 15 \rightarrow$ Poor consistency

Case A – $\sigma = 4$

For $\sigma = 4$:

Using empirical rule:

68% of bulbs last between:

$55 \pm 4 \rightarrow 51$ to 59 days

95% between:

$55 \pm 8 \rightarrow 47$ to 63 days

99.7% between:

$55 \pm 12 \rightarrow 43$ to 67 days

Interpretation:

- Almost all bulbs last near 55 days.
- Very few early failures.

- High customer trust.
- Excellent quality control.

Ideal for:

- High-end manufacturing
 - Brand reputation
 - Warranty-based products
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Case B – $\sigma = 10$

For $\sigma = 10$:

68% between:

45 to 65 days

95% between:

35 to 75 days

99.7% between:

25 to 85 days

Interpretation:

- Moderate variation.
- Some bulbs fail early.
- Some last very long.
- Mixed customer experience.

Suitable for:

- Mid-range products
 - Cost-based competitive markets
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Case C – $\sigma = 15$

For $\sigma = 15$:

68% between:

40 to 70 days

95% between:

25 to 85 days

99.7% between:

10 to 100 days

Interpretation:

- Very inconsistent production.
- Some bulbs fail extremely early.
- Some last unusually long.
- High warranty costs.
- Poor quality control.

This indicates:

- Production instability
 - High defect rate
 - Low reliability
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Graphical Interpretation

(Refer to the graph shown earlier)

Observations:

$\sigma = 4$:

- Tallest peak
- Narrowest spread
- Most reliable production

$\sigma = 10$:

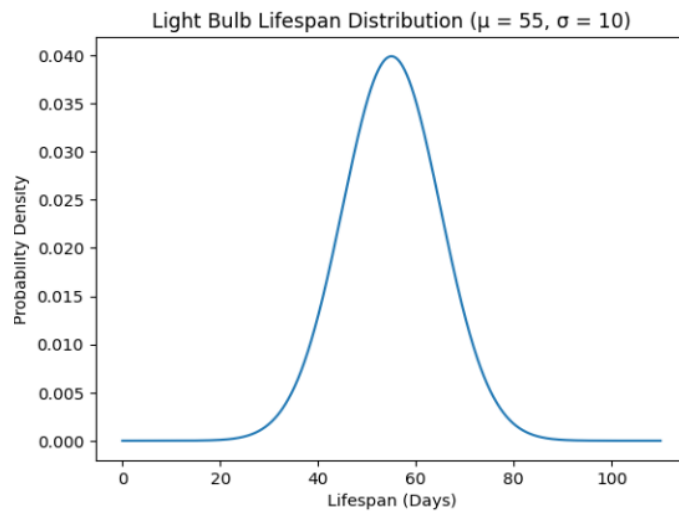
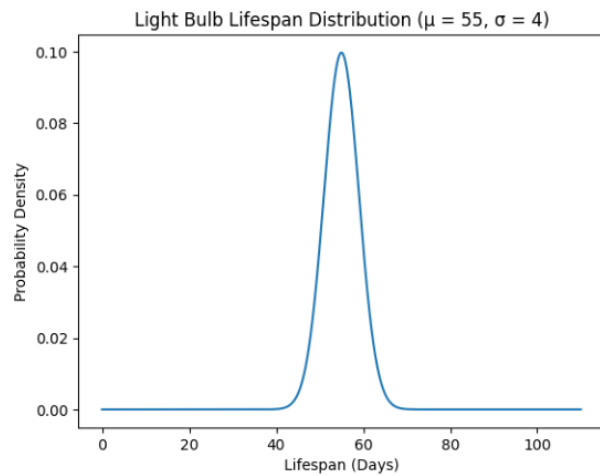
- Moderate spread
- Average consistency

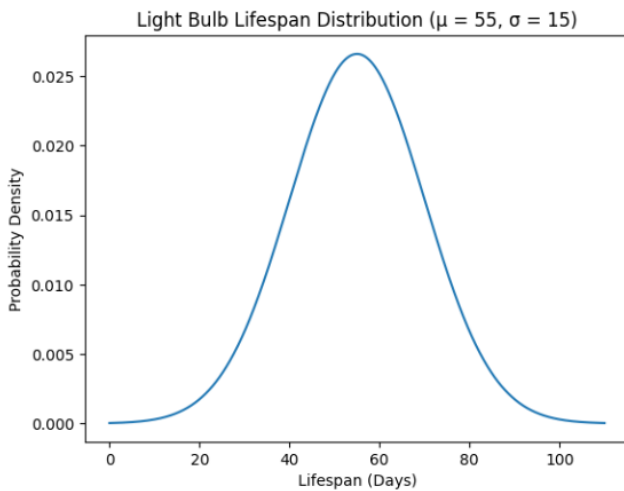
$\sigma = 15$:

- Widest spread
- Unstable manufacturing

As σ increases:

- Reliability decreases
- Customer risk increases
- Brand reputation decreases





Comparative Table

Case Mean (Days) Std Dev Quality Level Consistency Customer Satisfaction

A	55	4	Excellent	Very High	Very High
B	55	10	Moderate	Medium	Medium
C	55	15	Poor	Low	Low

Key Insight:

In manufacturing, **consistency is more important than average value.**

Even if average lifespan is 55 days, high variability can damage brand image.

Conclusion

Given:

$\mu = 55$ days

$\sigma_a = 4$

$\sigma_b = 10$

$\sigma_c = 15$

All three manufacturing processes produce bulbs with the same average lifespan.

However, variability differs.

Best Manufacturing Process:

✓ $\sigma = 4$

Because:

- Highest precision
- Least variability
- Lowest defect rate
- Most predictable performance
- Highest customer satisfaction

$\sigma = 10$ may be acceptable for mid-level markets.

$\sigma = 15$ is not recommended for quality-controlled industries.

Final Answer:

In industrial manufacturing and quality control, $\sigma = 4$ is clearly the best process.